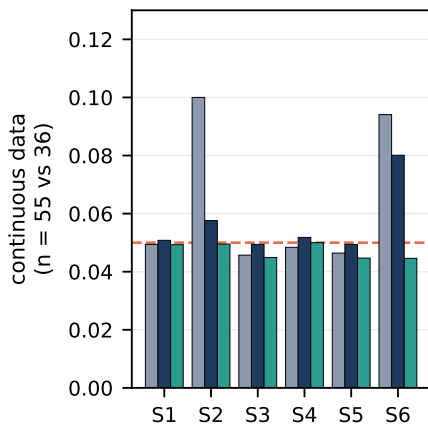
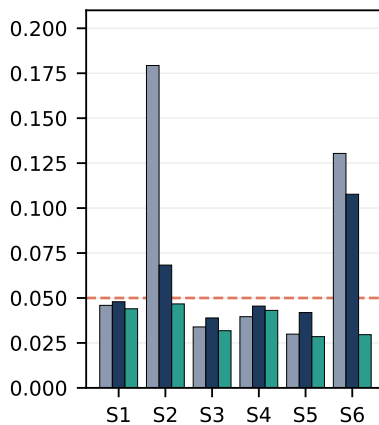


# Guardian cascade calibration: Type I error, FDR and power under known ground truth

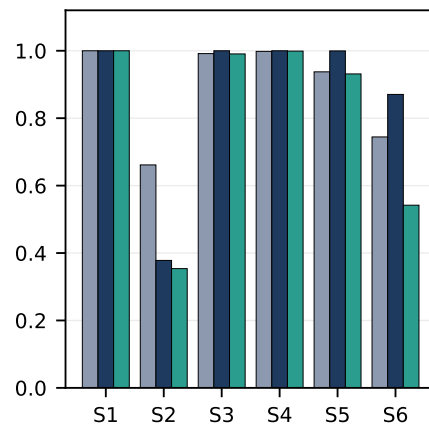
## A. Type I error (raw $p < 0.05$ )



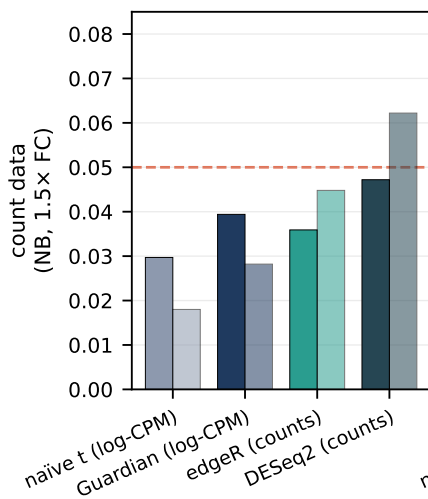
## B. FDR (BH $q < 0.05$ )



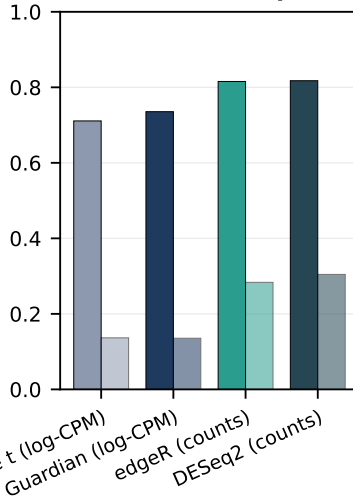
## C. Power



## D. Count data — FDR



## E. Count data — power



Bar colour / shading

- naive Student t (gate off)
- Guardian cascade
- always-Welch (reference)
- nominal  $\alpha = 0.05$
- D-E solid: 55 vs 36 samples
- D-E faded: 20 vs 20 samples

A-C scenario codes

- S1 normal, equal var.
- S2 normal, unequal var.
- S3 heavy-tailed (t, df 3)
- S4 lognormal (skewed)
- S5 outlier-contaminated
- S6 unequal var. + heavy